where K is the kinetic energy, and the mean auxiliary variable ξ

$$\bar{\xi}(t) = \int_{\mathcal{D}\times\mathbb{R}} \xi\rho(\boldsymbol{q},\boldsymbol{p},\xi,t)\mathrm{d}\omega\mathrm{d}\xi.$$

ρ that appears here is the normalized pre-equilibrium density which evolves according to the Fokker-Planck equation of the extended Nosé-Hoover-Langevin system.

It is then possible to perform a local stability analysis to characterize the convergence to equilibrium of $\bar{\theta}$ and $\bar{\xi}$ to the equilibrium $(\mathrm{N_d k_B T}, 0)$. Although the analysis makes certain stringent assumptions they are verified in [225] for a Lennard-Jones molecular system.

8.4.4 *Velocity Rescaling*

An alternative stochastic-dynamical method for thermostatting molecular dynamics is the stochastic velocity rescaling method proposed by Bussi and collaborators [60, 63] and somewhat generalized in [225]. The equations are (for the case $\boldsymbol{M} = \boldsymbol{I}$, considered for simplicity):

$$\begin{aligned}\mathrm{d}\boldsymbol{q} &= \boldsymbol{p}\mathrm{d}t\\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\,\mathrm{d}t - \Psi(K)\boldsymbol{p}\,\mathrm{d}t + \sqrt{2k_\mathrm{B}\mathrm{T}\Phi(K)}\boldsymbol{p}\mathrm{d}W,\end{aligned}$$

where $W(t)$ is a single Wiener process, Φ is an arbitrary smooth positive function and Ψ is defined by

$$\Psi(K) \stackrel{\mathrm{def}}{=} (2K - (1+\mathrm{N_d})k_\mathrm{B}T)\Phi(K) - 2k_\mathrm{B}\mathrm{T}K\frac{\mathrm{d}\Phi}{\mathrm{d}K}.$$

With these definitions, the method preserves the canonical probability density ρ_β. We do not prove the ergodicity of the method here, but expect that it would follow by an analysis similar to that given above for Nosé-Hoover-Langevin.

The original proposal of Bussi et al. had

$$\Phi(K) = \frac{\gamma_\mathrm{VR}}{2K}, \quad \text{so that} \quad \Psi(K) = (1 - (\mathrm{N_d}-1)k_\mathrm{B}\mathrm{T}/2K)\,\gamma_\mathrm{VR},$$

where γ_VR is a positive constant.

For the more general case, the required conditions on Φ are as follows:

$$K\Phi(K) \text{ is bounded as } K \to 0, \qquad \Phi(K) \text{ grows at most polynomially as } K \to \infty.$$